



Agentic Systems in Biomedicine

Workshop — an introduction to LLM-powered agents for biomedical research

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Outline

What we will cover in this introduction

1

Terminology & concepts

LLMs, agents, tools, multi-agent systems

2

How agents work

How an LLM-based agent thinks and acts

3

API keys explained

Authentication for connecting to model providers

4

Course agent lineup

The six agent types we will build together

5

Biomedical systems today

Real biomedical agent systems in production

6

Strengths & limits

Limitations, safety, and responsible deployment

Chatbot vs. Agent

The fundamental difference between answering and acting

Chatbot

Generates a text response based only on its training. One turn in, one turn out. It cannot access the internet, run code, or check a database — it can only describe what it would do.

Agent

Plans the steps, uses external tools, observes results, and decides what to do next. The LLM is the brain, but it now has hands — it can search, compute, query, and write files.

Bottom line: an agent is an LLM that can take actions in the world, not just describe them.

Key Terminology

An agent is built in layers

Agent — the orchestrator

Plans, decides, and runs the loop. Takes a goal, picks which tool to call next, reads the result, decides whether to keep going.

MCP — the standard wiring

Model Context Protocol. A common language for plugging tools into agents — like USB for AI. Connect once, reuse anywhere.

Tools — the hands

External functions the LLM can call. Web search, Python execution, BLAST, PubMed query — anything with a defined input and output.

LLM — the brain

Large Language Model. The reasoning engine — GPT, Claude, Gemini, Llama. Generates text and tool-call instructions.

How an Agent Works

A loop, not a pipeline — the agent keeps cycling until the goal is met



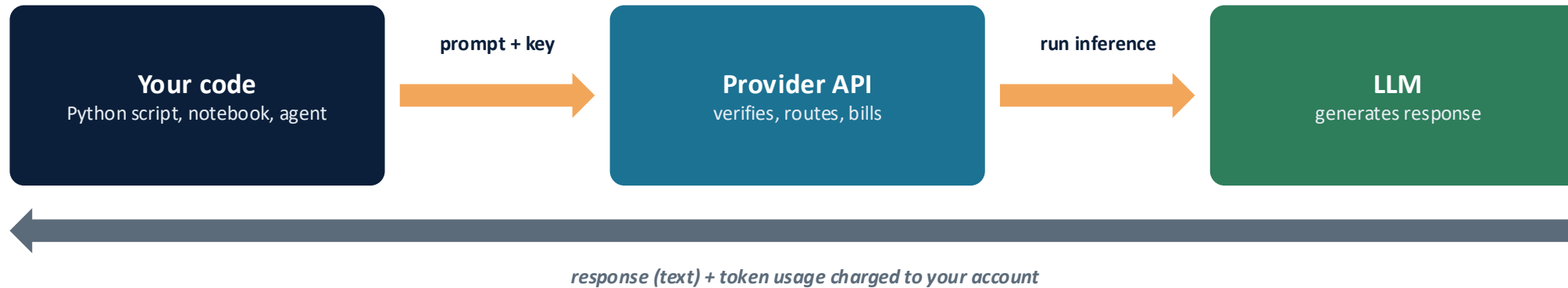
WALKTHROUGH: find genes linked to long-QT syndrome

- 1 Plan**
"I need disease-gene associations. Use OMIM."
- 2 Call Tool**
`omim_search("long QT syndrome")`
- 3 Observe**
→ KCNQ1, KCNH2, SCN5A, ...
- 4 Decide**
"Need pathway context → Loop with KEGG."

The loop may run 5 or 50 times before the agent reports a final answer.

What Is an API Key?

The credential that lets your code talk to a model provider



WHAT IT LOOKS LIKE IN CODE

```
# load from environment, never hardcode
api_key = os.environ["OPENAI_API_KEY"]

# the key is sent in every request header
client = OpenAI(api_key=api_key)
resp = client.chat.completions.create(
    model="gpt-4",
    messages=[...])
```

WHAT A KEY LOOKS LIKE

sk-proj-aB3xK9p2Lm4Q...Wn8z
~50 random characters — uniquely identifies you

⚠️ Treat it like a credit card

- Never commit to Git or paste in chats
- Store in env vars or a secrets manager
- Rotate immediately if leaked
- Leaked keys = stolen money



The Agents We Will Build

Chatbot Agent

- The familiar entry point: a conversational interface
- Adds tool calls — web search, calculator, code execution
- Your first hands-on exercise of the workshop

Personal Assistant

- Reads your calendar, drafts emails, manages notes
- Connects to multiple services through MCP
- Shows how agents coordinate across everyday tools

Lab Interpreter

- Loads your experimental data and runs analysis
- Writes Python on the fly: stats, plots, summaries
- Turns raw CSVs into publishable figures

Research Agents

- Deep Research: multi-step web investigation across sources
- Literature Agent: PubMed search, paper summarization, citation graphs
- Both demonstrate long-horizon, autonomous reasoning

Plus: and a live demo of an HPC Operator agent that submits jobs to a compute cluster on your behalf.

Biomedical Agents

Three production-grade agentic systems published in 2024–2025

Biomni

Stanford, 2025

A general-purpose biomedical agent that autonomously runs research tasks across genomics, drug discovery, and clinical analysis.

How:

- ▶ Causal gene prioritization
- ▶ Drug repurposing & rare-disease
- ▶ Multi-omics (wearables, scRNA-seq)

- + Broadest scope, runs locally with your data
- Executes code with full privileges — sandbox required

biomni.stanford.edu

TxAgent

Harvard, 2025

Therapeutic-reasoning agent that analyzes drug interactions and contraindications using 211 curated biomedical tools.

How:

- ▶ FDA drug database integration
- ▶ Open Targets & Monarch ontology
- ▶ Multi-step clinical reasoning

- + Grounded in real, up-to-date drug data
- Narrow to therapeutics; limited beyond drugs

github.com/mims-harvard/TxAgent

ChemCrow

EPFL, 2024

Chemistry agent that plans organic synthesis, designs molecules, and queries reaction databases through 18 expert tools.

How:

- ▶ Synthesis planning & retrosynthesis
- ▶ Molecule property prediction
- ▶ Reaction execution on cloud labs

- + Tight integration with lab workflows
- Chemistry-only scope; needs lab access for full power

github.com/ur-whitelab/chemcrow

What to Watch Out For

The principles for safe and meaningful agent deployment in research



Real limitations

Agents hallucinate, make wrong tool choices, can leak credentials, and execute code they should not. In biomedicine, a confident-but-wrong answer can mislead an entire experiment.



Safe practice

Sandbox code execution, keep humans in the loop for critical decisions, log every tool call, and treat agent outputs as hypotheses to verify — not conclusions to trust.

Bottom line: an agent is a powerful collaborator, not an oracle. Verify before you trust.



What to Take Away

The mental model

- Chatbot answers; agent acts
- Goal → Plan → Tool → Observe → Decide
- The loop runs until the goal is met

The vocabulary

- LLM, Tool, Agent, MCP
- API key — your authenticated access
- Multi-agent — agents that call agents

The applications

- Biomni for general biomedical research
- TxAgent for therapeutic reasoning
- ChemCrow for chemistry workflows

The mindset

- Sandbox everything that runs code
- Verify outputs — do not trust blindly
- Keep a human in the loop for critical steps

Next: we build our first agent — starting from a chatbot and adding tools, one at a time.